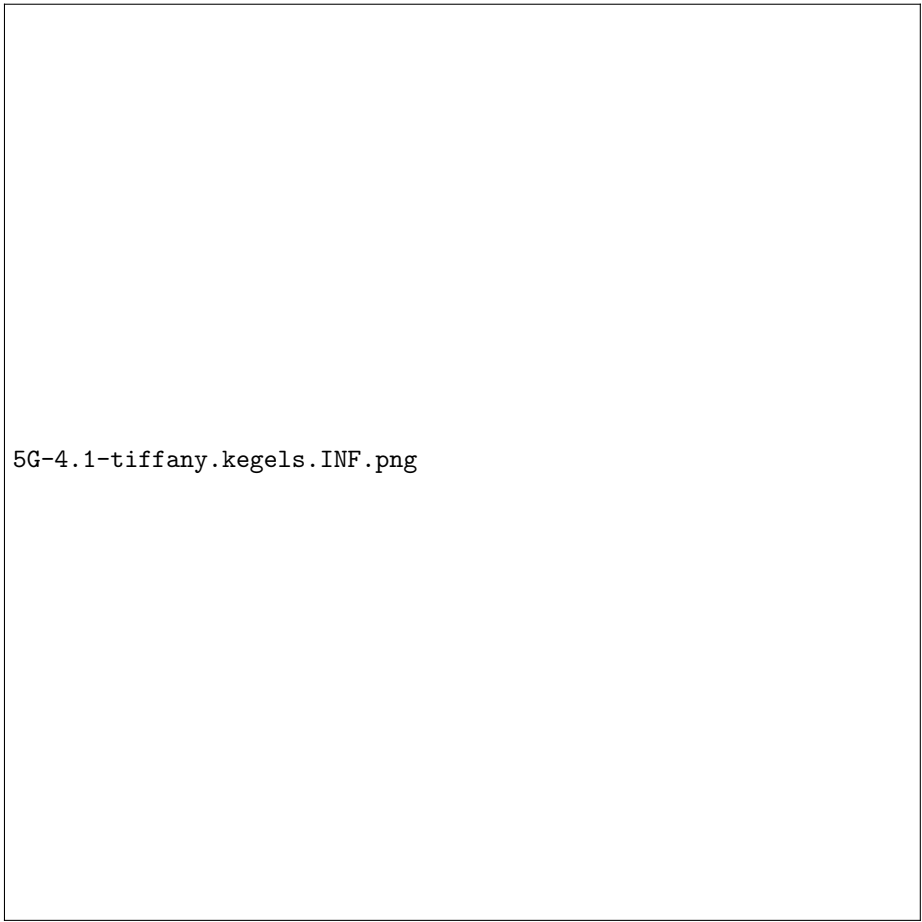


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Oefeningenreeks 5G nr. 4.1

$$\begin{aligned} V &= \pi \int_0^{\frac{\pi}{4}} \cos^2(y) \, dy \\ &= \pi \int_0^{\frac{\pi}{4}} \frac{1 + \cos(2y)}{2} \, dy \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{4}} (1 + \cos(2y)) \, dy \\ &= \frac{\pi}{2} \left[ y + \frac{\sin(2y)}{2} \right]_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{2} \left( \frac{\pi}{4} + \frac{\sin(\pi/2)}{2} \right) \\ &= \frac{\pi}{2} \left( \frac{\pi}{4} + \frac{1}{2} \right) \\ &= \frac{\pi^2}{8} + \frac{\pi}{4} \end{aligned}$$

## References



5G-4.1-tiffany.kegels.INF.png